Worksheet Solution: Graphs of physical processes

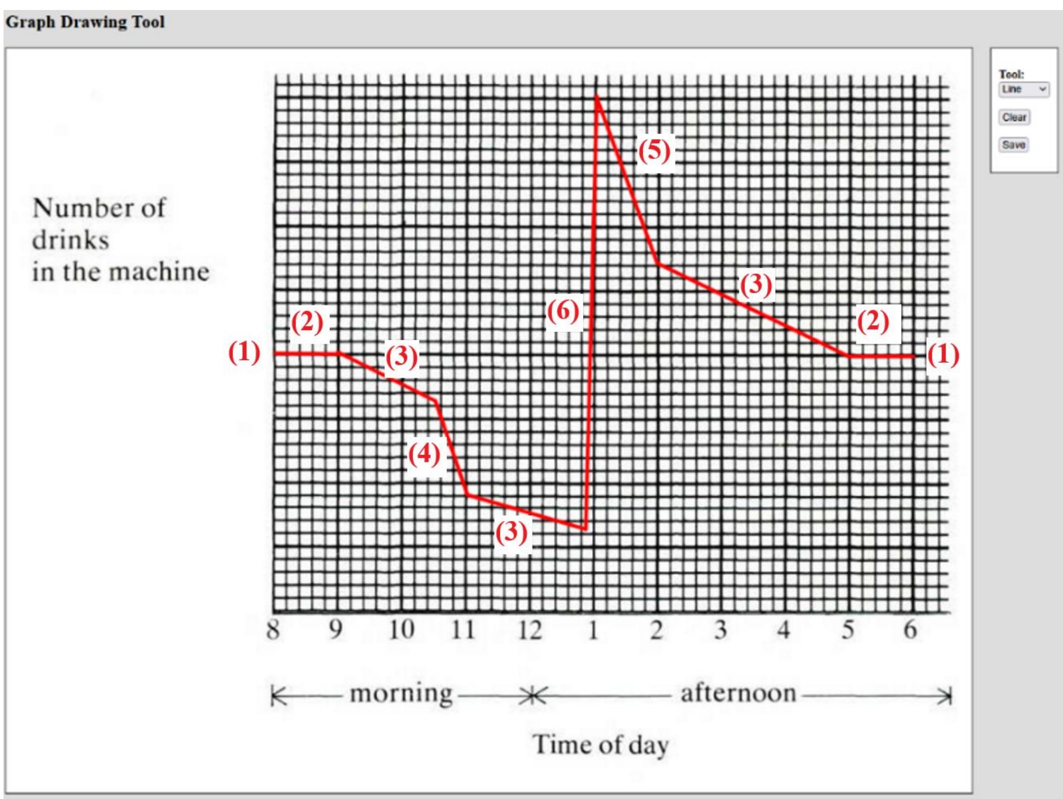
Part I: Cafeteria vending machine

A cafeteria has a vending machine selling sodas. Breakdown:

(1) On a typical day, the machine starts half full.
(2) No drinks are sold before 9AM and after 5PM.
(3) Sodas sell at a slow rate throughout the day ...
(4) ... except during the morning from 10:30AM to 11AM and
(5) ... at lunch break from 1PM to 2PM.
(6) The machine is filled once during the day right before the lunch break. It takes approximately 10 minutes to fill the machine.

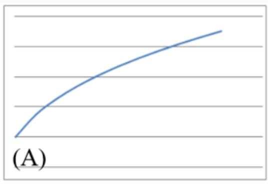
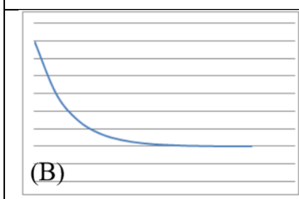
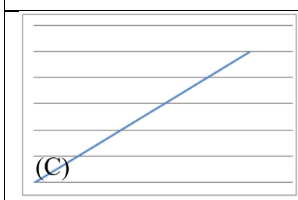
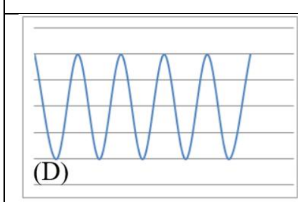
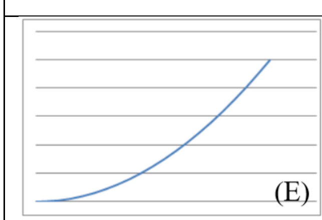
1. Sketch a graph showing the amount of soda cans inside the vending machine at different times on a typical day below

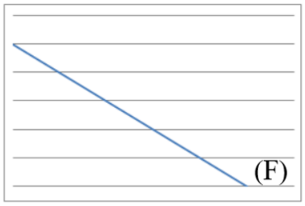
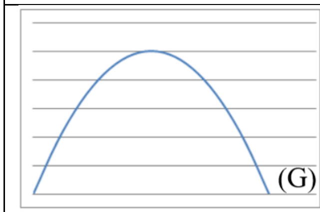
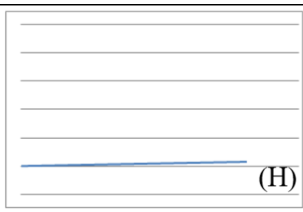
Solution with corresponding descriptions labelled



Part II: Name that graph!

These are graphs of some physical processes

Graph	Process	Formula	Reasoning
	(7) I blow up a roughly spherical balloon using a balloon pump. After each pump, I measure the radius of the balloon in centimetres. What would the radius-number of pumps graph look like?	$y = A\sqrt[3]{x}$	Each pump adds roughly same volume to the balloon. To get the radius, we recall volume formula for sphere ...
	(1) A cup of tea is made and temperature measured in deg C every second. What would temperature-time graph look like?	$y = \frac{A}{e^x} + B$	Newton's law of cooling ...
	(4) I measure several objects using inches and then using metres, plot them on a scatter graph, and join the points. What would the metres-inches graph look like?	$y = Ax$	Linear relation. 0 metre is 0 inches; therefore, the y-intercept is 0.
	(3) The height of a valve on a bicycle tyre above the ground is measured after each centimetre that the bicycle travels forward. What would the height-distance graph look like?	$y = A\sin(Bx) + A$	Note: this is the only graph that is oscillating and periodic. The actual trajectory is a cycloid.
	(5) I jump out of a plane and the distance fallen from the plain is measured every 0.1 sec until I open my parachute. What would distance fallen-time graph look like whilst in freefall?	$y = Ax^2$	Constant acceleration in free fall. Note: distance fallen is not height!

	(8) I suck water through a straw out of a large beaker at a constant rate and measure the volume of liquid remaining at various times. What would the volume-time graph look like?	$y = Ax + B$	Constant rate of decrease. Linear function with negative slope.
	(3) I throw a tennis ball straight up into the air and catch it. The height of the ball from the ground is measured over the time of the journey using freeze-frame photography. What would height-time graph look like?	$y = Ax^2 + Bx + C$	Constant negative acceleration with positive initial velocity …
	(6) I drive 70 miles an hour along a motorway and note the reading on my odometer (mile counter) every 5 minutes. What would the odometer reading-time graph look like?	$y = Ax + B$	Constant increase, linear function. Slope is small because odometer initial value is likely already large!